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Segmenting the target audience for transportation demand management programs: An investigation between mode shift and individual characteristics

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ABSTRACT
Understanding the characteristics of travelers and situations that are more likely to switch from single occupancy vehicles (SOV) to more sustainable mobility options can help transportation agencies develop better behavior change strategies. Existing mode shift programs, however, rarely identify their target audience, which could make the programs less effective. To investigate the demographics and trip characteristics of those who are more likely to shift from SOV to carpooling, public transit, and micromobility (e.g. walking, biking), a matrix decomposition audience segmentation method is proposed and applied on travel survey data. The results of this research show that for short-distance leisure or social trips, young and middle-income people are more likely to shift to these mobility options from SOV. This study provides a comprehensive and profound understanding of the characteristics of the target audience that can inform policies to create more targeted behavior change strategies to reach their sustainability goals.

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1. Introduction

Transport and mobility is one of the sectors targeted where effective interventions are being called for to reduce CO2 emissions and where adaptation measures are needed to reduce the vulnerability to climatic change (Stead, 2016; Zhu et al., 2022). The mission of sustainable mobility is to provide services and infrastructure for the safe, cheap, accessible, efficient, and resilient movement of people and commodities while reducing carbon and other emissions and environmental impact (Mahmoud Mohieldin, 2017). A move from single occupancy vehicles to more sustainable mobility options, such as public transit, carpooling, and micromobility, is critical to reducing global emissions from the transportation sector. Changing traveler behaviors and habits is needed to accomplish this goal.

Transportation demand management (TDM) programs aim to help people know about and use all their mobility options such as transit, carpooling, walking, and biking. TDM programs have historically targeted commuter trips and employers (Cooper, 2007). More recently, TDM programs at the household level and for all types of trips are being approached through the lens of social and individualized marketing, not only for employers, but for state and regional agencies, primary schools, universities, communities, and special events (Dill et al., 2010). Aided by advancements in smartphone technology, some transportation agencies have begun to develop apps to provide incentives to trigger individual travel behavior change, such as a mode shift (T. Li et al., 2021). Identifying the target audience prior to rolling out a program can help reduce the cost of the program, including program development, delivery, and/or marketing, given the larger size of the prospective audience party originally needed, prior to target audience identification. Furthermore, it would be challenging to create effective individualized marketing campaigns without a solid understanding of the target audience's characteristics.

For transportation agencies to understand who might (or might not) want to utilize these alternatives, an analysis of the characteristics of those who are more (or less) likely to switch from their personal vehicles to different alternatives, is necessary. Focusing on developing TDM strategies for a set of the target audience who are more likely to change allows transportation professionals to fully appreciate the potential of these strategies and to understand which group(s) they will likely have the greatest effect on (Boslaugh et al., 2005). Knowing the target audience profiles of the TDM programs allows the audience's interests and requirements to be prioritized in the marketing strategy to ensure they follow the next step toward their behavior change. Furthermore, knowing how many various types of target audiences could be included in these programs can aid in the development of behavior change strategies for them. Previous research has examined target groups in order to provide targeted messages to encourage changes in a variety of socially important behaviors such as drug use and smoking (McClure et al., 2006; McDonald, 1999; Raw et al., 1998; Schmid et al., 2008). Because it is impractical to reach everyone at once, identifying and

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segmenting target audiences is critical for ensuring that marketing and behavior change strategies are targeted and not wasted on the wrong people.

We addressed the research gap in understanding the target audience of TDM programs by identifying the characteristics of people who are most likely (and least likely) to switch to carpooling, micromobility, or public transit. In particular, we aimed to identify people who currently primarily rely on their personal vehicle as their best mode of transportation but who are also motivated (e.g. would like to try) and capable of using alternatives (e.g. have access to, have the time flexibility to utilize). This propensity for the alternative could be a sign that they are considering adopting them but have not yet taken action because they always have their own vehicles available, among other reasons. According to the Motivation Theory by Dr. BJ Fogg at Stanford University (Fogg, 2009), if a change is easy to do and an individual is motivated to change, there is a greater chance for them to engage in a new behavior when they are triggered or encouraged to do it. In this study, we, therefore, defined the target audience after taking into account both motivation and capacity in terms of mode shift. Participants were specifically asked to choose their best and second-best mobility options depending on both their motivation and capability.

Understanding the motivated and capable population is critical to help future transportation demand management programs identify the audience that is most likely to switch to another mode of transportation, maximizing the impact of a TDM program (Hunecke et al., 2010). In this study, we also characterized those who lack the motivation or capability to determine the type of audience that transportation agencies would not wish to target. To segment people based on their characteristics and the mode they are willing to switch to, we proposed a non-negative matrix segmentation approach and developed personas for those who are likely to shift to public transit, micromobility, carpool, and not likely to shift. We also examined and compared the characteristics of different personas, such as trip characteristics, demographics, and lifestyles.

Overall, we developed a framework for segmenting the target audience who are more likely to change their mode of transportation from driving to more sustainable modes of transportation for transportation demand management programs. This research presents two primary contributions. To begin, state and regional agencies with sustainability goals can employ the target audience characteristics and develop an analysis framework/method to select the target audience for marketing objectives to be used in their own region. Second, the characteristics of the target audience can facilitate the development of more targeted TDM programs when designing for sustainable behavior change, which is imperative to consider for the adoption of sustainable modes for different demographic characteristics and individual circumstances.

2. Literature review

The concept of a target audience is frequently utilized in marketing and personalization. The target audience is a defined set of people who are most likely to use the product or service, and to whom marketing efforts should be directed. Age, gender, wealth, geography, interests, and a variety of other criteria can all influence who the target audience is. This section mainly summarizes the past literature on TDM programs, target audience segmentation applications, influencing factors of mode shift motivations, and methods used for audience segmentation.

2.1. TDM programs and target audience segmentation

TDM research and programs aim to comprehend behavior and the tradeoffs people make when choosing between sustainable mobility options (e.g. biking, public transit) and driving single-occupancy vehicles (SOVs). They also develop efficient behavior change strategies to initiate and sustain behavior change that lowers vehicle miles traveled and greenhouse gas emissions. Informational campaigns, behavioral interventions that affect people's attitudes or norms, monetary or non-monetary incentives, congestion or parking pricing, and gamification are some of the strategies used in these programs (Alta Planning + Design & Behavioral Insights Team (BIT), 2018; Dahlinger et al., 2018; Kazhamiakin et al., 2015). The impact of these programs include mode shift toward more sustainable mobility options, congestion mitigation, and reduction in greenhouse gas emissions (Federal Highway Administration, 2018; Greene-Roesel et al., 2018).

TDM programs come in a variety of types. TDM strategies have been effectively applied in a variety of urban settings, frequently in metropolitan public universities and large companies. Because of their concentrated administrative resources and focused target audience, these employers have been seen as the best places to test and implement TDM programs (Riggs, 2015). In order to promote non-car modes of transportation, including but not limited to employees, university employees and students, or commuting trips specifically, social marketing is becoming an increasingly popular TDM strategy. A detailed understanding of the needs, desires, and perceived barriers of the target audience is the cornerstone of effective social marketing. It employs a variety of techniques to produce an appealing offering that is tailored to these needs, desires, and perceived barriers (Thøgersen, 2014). The platforms for performing social marketing include social media, smartphone apps, and/or websites (Hu et al., 2017). Additionally, some universities have also adopted social marketing TDM programs that encourage faculty, staff, and students to voluntarily change their mode of transportation while they can also earn incentives.

There are many discussions in regards to the methods to increase the cost-effectiveness of social marketing TDM campaigns (Winters et al., 2018). According to a thorough analysis of TDM program effectiveness by the Federal Highway Administration, TDM can only be highly effective at a relatively low cost (as opposed to capacity enhancements) when used in the proper location(s), at the proper time(s), and for the appropriate selection of a target market of travelers (FHWA Office of Operations, 2020). While

TDM programs encourage as much behavior change from everyone, not all travelers are good targets for TDM programs. Past programs have attempted to motivate everyone, including those with extremely low motivation, and engage them in mode shift tasks that then failed to establish or form permanent habits of using the alternatives (Cellina et al., 2019; Halvorsen et al., 2020). This raises the issue of how we can induce more people to change their behavior while also lowering the expense of TDM programs in activities such as marketing. To facilitate the social marketing approach to promote sustainable behavior, previous research has presented the idea of identifying several target groups that describe the heterogeneous motivations and needs of all consumers (Hunecke et al., 2010). For instance, research has shown that those who care more about environmental issues are more receptive to TDM programs (Doran et al., 2017).

The benefits of using audience segmentation when developing TDM programs can be demonstrated by several effective initiatives to cut back on peak-hour transit use. One TDM program deployed on the Mass Transit Railway (MTR) network in Hong Kong used incentives to encourage users to change their departure times to lessen congestion during rush hours. While over 27% of adults commuting in the morning peak period to qualifying stations received a discount for exiting in the pre-peak hour, only approximately 2.5% of the morning trips actually shifted from the peak. Therefore, the authors further examined the characteristics of these trips and found that when trips are longer, users are less likely to respond to the offer. Furthermore, the socio-economic and employment characteristics of users who live outside of the city center may influence the possibility of them using the system (Halvorsen et al., 2020). The authors then employed a customer segmentation approach to better understand the diverse responses of different groups. Users with a lot of schedule flexibility and experience with the system, as well as those who travel vast distances, were more likely to adjust their travel dates to take advantage of the discount. This longitudinal promotion research suggests that 35–40% of passengers who first adopted the promotion would eventually revert to their former travel time periods (Z. Ma et al., 2020).

Although these examples present some benefits of an audience segmentation approach in departure time shift TDM programs, it is not well understood how we can identify and segment the target population to support mode shift programs. As a result, in this study, we focused on identifying the people who have the capacity and motivation to switch from personal vehicles to alternative modes such as public transit, micromobility, and carpooling. Research has shown that marketing campaigns and interventions have a higher chance of success if they are prioritized based on the target audience characteristics.

Target audience identification has been employed in many domains such as brand marketing, health care, personalization, and promoting pro-environmental behavior including transportation mode shift behavior (dos Reis et al., 2022; Gallagher et al., 2020; Lo et al., 2015; Schmid et al., 2008). Audience segmentation approaches have been extensively investigated in order to build more effective messages regarding mitigating or adapting to climate change threats. One of the longest-running segmentation programs was conducted by the Yale Program on Climate Change Communication and the George Mason University Center for Climate Change Communication. The survey used for audience segmentation was conducted from June 2017 to March 2021. Multiple variables assessing climate change beliefs, issue involvement, policy preference, and behavioral responses were utilized to segment a nationally representative sample of U.S. residents. Six distinctive segments emerged which rejected quantitative shifts from generally high to generally low levels of concern, issue engagement, and degree of certainty that global warming was occurring: Alarmed, Concerned, Cautious, Disengaged, Doubtful, and Dismissive ("Segmenting the Climate Change Alarmed: Active, Willing, and Inactive," n.d.). Using 36 variables on global warming beliefs, risk perceptions, policy preferences, and behaviors, Latent Class Analysis (LCA) was employed to identify the six segments that produced the most distinct and interpretable groups. More details about the methodology used in the target audience segmentation for this study are presented in Section 2.2.

A large portion of current transportation segmentation analysis research is devoted to determining the characteristics of people who travel by different modes, which helps different service providers and traffic demand management programs identify their major markets based on geolocations, audience demographics, and other factors. One type of research focuses on audience segmentation based on their current mode choice. One study modeled travelers' mode preference to be a function of the decision-making environment and discovered six segments based on the Bay Area Travel Survey they defined as "Multimodals," "Empty nesters," "Transit takers," "Inveterate drivers," "Moms in cars," and "Car commuters" (Vij & Walker, 2014). Another study looked into the characteristics of people who are inclined to utilize Uber or Lyft (Alemi et al., 2018). Based on people's lifestyles and life stages, three types of audiences were identified, among which the results showed that millennials who are well-educated and self-sufficient (i.e. who have already established their household), and those who live in high-quality transit neighborhoods are more likely to utilize Uber or Lyft.

2.2. Influencing factors of mode shift motivation and behavior

In order to group travelers into potential 'mode switchers,' one study examined psychological aspects such as intentions, habits, environmental worldview, and knowledge. The authors discovered that people who were described as "Aspiring Environmentalists" (the most educated category of people), had the strongest intention to switch from driving to alternative modes of transportation. The study also pointed out that raising public awareness of a car's harmful environmental effects is rarely enough to change behavior, as many identified groups have either unfavorable attitudes toward alternatives or a lack of moral responsibility for changing their behavior (Anable, 2005). One study further examined the likelihood to shift mode and location factors and found that density and transit supply are positively associated with the potential transit use (Brown et al., 2016). Another study used sociodemographic characteristics and behavioral parameters such as the number of trips per day and week, as well as departure time trends, to divide travelers into groups depending on their proclivity to change their departure time (avoid travel during peak hours). People described in their study as "Road Warriors," those with the most trips (average number of daily trips was 4.5), were found to be more inclined to adjust their departure time. Furthermore, the study used a smartphone-based experiment to see if some segments are more inclined than others to change their departure time. However, the experiment was conducted with only 24 people. Larger sample size would likely be more effective in demonstrating that the segmentation strategy was helpful in generating diverse audiences to facilitate communications (Arian et al., 2021).

Other studies have used a larger range of psychological factors, demographics, and lifestyles to better understand distinct groups of behaviors, beliefs, or attitudes regarding pro-environmental measures like climate change and energy saving (Line et al., 2012; Musti et al., 2011; Saleem et al., 2018). A study suggested that more perspectives might be needed to produce different audience segments based on distinct lifestyles such as career lifestyle, outdoor lifestyle, and pro-environmental lifestyle (Axsen et al., 2012).

There are not enough studies on combining these factors to generate a set of profiles that characterize people who are more likely to switch from their personal vehicles to more sustainable alternatives, despite the fact that the influential factors of mode shift behavior have significant implications for developing TDM programs, according to the previous literature. With the help of the target audience profiles, transportation agencies can build more tailored TDM programs for a given group or groups of people by having a clearer picture of what the target audience looks like. In this study, we addressed this research gap by using a matrix factorization approach to create personas of those who are more inclined to switch to public transportation, micromobility, or carpooling using demographics and activity patterns. The results of the different personas and the audience approach can help deepen the understanding of the target audience types for future TDM programs.

2.3. Methods of audience segmentation

Individuals that display similar patterns of responses across a collection of variables are identified using segmentation algorithms. One of the most frequent ways of audience segmentation is clustering. Clustering algorithms reduce within-group disparities while maximizing differences between groups. K-means clustering has been used to divide travelers into different groups with diverse travel behaviors and sensitivity to incentives (Arian et al., 2021; M. Lee et al., 2022; Mendiate et al., 2020). Another clustering algorithm called hierarchical clustering was used to cluster a list of users who are followers of AJ+ (an online news channel) Facebook page based on their gender, age, country, and the URL they shared (An et al., 2017). The benefit of hierarchical clustering is that it produces a tree-based representation containing a nested sequence of clusters without requiring the number of clusters to be input or prior knowledge. As a result, it is suitable for situations in which the number of clusters needs to be adjusted based on user input and clusters need to be found on demand. It is, however, more vulnerable to outliers than the K-Means algorithm, making it less robust when new data is introduced (B, 2020).

Another study proposed a non-negative matrix factorization (NMF) approach and compared it with K-Means to cluster the users who were YouTube channel followers based on their age, gender, and their liked videos (An et al., 2018). They found that NMF could detect more segments of users while K-Means tended to cluster many types of users into one segment, which showed that NMF can provide more meaningful insight into the heterogeneity of the audience. It has also been found to be more effective than principal component analysis (PCA) in revealing latent clusters since it leads to a part-based decomposition of the data and increases multivariate data clustering capabilities (Han & Moutarde, 2012). This is mostly owing to its non-negative limitations for creating cluster coefficient matrices, whereas PCA may generate matrices with both positive and negative values, making the clustering results difficult to comprehend (Namgung & Jun, 2019). NMF was also applied to segmenting gamer players based on gender, age, and preferences in gameplay elements based on the data from an online survey (Salminen et al., 2020). The player personas were summarized and could be used to suggest games to players depending on which persona they most closely resemble. However, the information used to create the personas is limited to merely age and gender, leaving the gamers with a limited representation.

Another common approach is latent class analysis (LCA) and its variations. It is a statistical method for identifying unmeasured class membership among subjects. The main difference between LCA and clustering methods is that the former is a top-down approach (starting with describing the distribution of data) while the latter are bottom-up approach (identifying similarities between cases) (Stephanie, 2015; Weller et al., 2020). The LCA algorithm implies that the data is made up of many distributions, each of which represents a latent class, or cluster, whereas the clustering algorithm makes no assumptions about the data. A study, for example, used a class membership model to uncover the latent classes that influence passengers' mode selection. Travelers' mode decision processes can be characterized by multinomial logistic models for each latent class, albeit with various parameters (Y. Lee et al., 2020). This modeling technique may not be beneficial to identify latent classes when the connection between the dependent variable (e.g. tendency to change mode) and independent factors (e.g. demographics) is unknown.

Nonnegative matrix factorization was utilized in this study to segment the audience based on their demographics and trip activities, based on a review of the methodology. Because of its potential to yield meaningful and interpretable results, it is regarded as a suitable methodology in this study compared to the other considered analytical approaches.

In terms of data collection to conduct audience segmentation, surveying is one of the most widely used ways to collect data about people's preferences, demographics, or other factors to conduct segmentation analysis. One survey study analyzed the Bay Area Travel Survey, which included mode preferences and trip data, in order to better understand the types of travelers in terms of mode choice (Vij & Walker, 2014). This study demonstrates that surveys are an effective technique for gathering information on attitudes and preferences. Thus, we used surveying as the primary tool in this study to better understand the characteristics of the target audience because it is more effective at eliciting attitudinal information and preferences for sustainable behavior change in a more straightforward manner.

3. Data description

In order to understand the characteristics of the target audience of TDM programs, data such as demographics and motivation or actual mode shift behavior should be collected for analysis. For employer-based TDM programs, demographics and mode shift behavior can be easily obtained and used to understand who could be the target audience for future more tailored TDM programs. For other app-based or web-based TDM programs, some studies performed pilot studies to understand users' responses such as compliance with incentives (Arian et al., 2021). The surveying method was utilized when there were not enough tools to support data collection (Logan et al., 2020).

In this study, we designed a survey to understand the characteristics of the target audience. Surveying is a relatively efficient way to obtain samples with acceptable levels of accuracy compared with others such as focus groups and has been used in many previous audience segmentation studies (Chryst et al., 2018; Ihm & Lee, 2021).

The case study was in the Greater Los Angeles Region, California. Greater Los Angeles is the second-largest metropolitan region in the United States with a population of 18.7 million as of 2019 (US Census Bureau, 2021, pp. 2010–2019). They have one of the strongest public transit systems in the country, with subways, light rail, buses, and shuttles serving most of the areas in this region.

We used Amazon Mechanical Turk (MTurk) for distributing the surveys as it is a prominent platform that can collect responses swiftly. It allows users to publish survey links and target specific geographic areas, allowing only residents of those areas to view the post and complete the survey. We distributed the survey to the residents of Los Angeles County *via* the screening criteria on MTurk, the application area for this study in October 2021, and lasted for a month to reach the effective sample size (i.e. 385 respondents). On June 15, 2021, the State of California's reopening plan went into effect, ending constraints such as physical separation and capacity limitations. California lifted its mandate that requires people to wear masks and advised citizens to follow CDC recommendations, according to which people should wear masks in any indoor environment, including on trains, aircraft, buses, and ride-sharing services, as well as at transportation hubs (California State Government, 2022). According to the APTA Ridership Trends, transit ridership has been increasing since March 2021, with the more widespread availability of COVID-19 vaccines, the return of more people to offices, and the start of the increase of other types of outside-the-home activities (The American Public Transportation Association, 2022). Through December 2021, ridership remained consistent at 58% of pre-pandemic levels. Even though the COVID restrictions in the application region were relaxed when the data collection activities began, the survey results may still understate the population who are willing to switch to shared modes of transportation such as public transit due to the disease's prevalence. It is also worth noting that the data collection for this study came to an end before the Omicron variant spike at the end of December 2021, which resulted in decreased travel, workforce, and services in public transit.

We calculated the effective sample size based on the confidence level (set as 95%), the margin of error (set as 5%), and the degree of variability (set as 50%). For example, if the actual proportion of the target audience is 40%, then the estimated proportion from the survey would be between 40%-5% and 40%+5% with a 95% confidence level. The degree of variability is set as 50% to be more conservative, as it assumes a greater level of variability in the survey than 20% or 80% which would require a higher number of samples. The formulation is in (GD Israel, 1992). In total, 641 responses were collected. After identifying the inconsistencies among the answers to various questions and removing respondents who provided 'Prefer Not to Answer' in at least one question, a total of 464 valid responses were collected.

3.1. Survey design

In the survey, respondents were asked to consider trips taken with high frequency and to provide information about their top three most frequent trips. For each of the most frequent trips, respondents would answer questions on trip characteristics, mobility options preferences (the best and the second-best mobility option. This method aims to capture the most frequent trips that are representative of their daily travel behavior and gather some insight into their lifestyle, as people's frequent travel patterns can provide valuable insight into their lifestyles, which can affect their mode choice (Thøgersen, 2018). In addition, we asked the respondents about other travel-related attributes such as bike ownership and socio-demographic characteristics.

The survey has four sections and the structure and question content are outlined below. The first three sections of the survey asked respondents to think about their top three most frequent trips and then answer questions about each of these trips, while the last section asks respondents to provide information on other pertinent travel and socio-demographic characteristics. The survey is summarized below in the following paragraphs.

3.1.1. Trip characteristics

The survey asked about the trip characteristics of people's frequently made trips, which can influence people's choice of alternative modes according to the literature. For instance, a study completed in Los Angeles, California, found that carpooling and commute distance are positively correlated. Public transit use is more prevalent among university students who are near their friends and classmates (Zhou, 2012). Another study asked people about their trip purposes, departure time, and travel time in order to better understand the factors that influence their mode choice and their capacity to reduce carbon emissions (Yang et al., 2018). Based on the findings from these studies, we included the following trip-related questions.

- Trip purpose
- Frequency of the trip
- Time of day the trip usually takes place
- Day of week the trip usually takes place
- Level of flexibility in terms of departure time (leave early or late)
- Sociability: if the respondent takes the trip alone or with others
- Trip duration: the length of time it takes to make the trip
- Activity duration: the length of time staying at the destination of the trip

3.1.2. Mobility options preference

Prior research primarily employed surveys to explicitly ask about people's motivations or favor scores of a particular alternative to understand people's modal shift motivation and behavior (X. Ma et al., 2020; Rodriguez-Valencia et al., 2021). This study focused primarily on those who currently use their personal vehicle as their best mobility option but have a "second-best" mobility option based on their motivation (i.e. would like to try an alternative mode) and capability (i.e. are able to use an alternative mode). In the past, studies on mode choice analysis primarily focused on one's best option. Some studies asked participants' second-best option(s) to understand their mode switching choice. One study, for instance, asked respondents about their second-best choice in order to determine whether they intended to use a newly built transit infrastructure in the future (Ben-Akiva & Morikawa, 1990). As a result, we asked participants in this study about both their best and second-best option depending on their own capability and motivation. The following bullet points list the two survey questions that we asked.

- What is your best option for the trip that you make the most frequently based on your capability (i.e., able to access and use) and motivation (i.e., want to use)?
- What is your second-best option for the trip that you make the most frequently based on capability (i.e., able to access and use) and motivation (i.e., want to use)?

The first question helps identify those who are currently driving (i.e. those who select their best option as personal vehicle). The second question helps identify the modes people are willing to try. The participants were asked to choose only one option for each question. The set of mobility options and their definitions provided to the respondents includes the following. Carshare and rideshare services were also provided as alternatives, but they were not of interest in this study, as we focus on sustainable mobility alternatives. Vanpool was also provided as an alternative, but the number of respondents selecting this option was very low, thus not of interest in this study.

- Personal Vehicle: Drive alone
- Carpool: Traveling in your or other vehicles with 1-3 other people
- Public Transit: Such as rail or bus
- Micromobility: Walk, bike, scooter, or other shared modes
- None: In terms of their second-best mobility option, participants got the option to select 'None,' indicating that they did not have a second-best option either due to lack of motivation or not being able to use.

3.1.3. Bike ownership

We asked the respondents how many bikes (owned or shared) their household owns or regularly uses. In addition, the study also asked about vehicle ownership. It had no significant impact on segmenting the target audience, hence it was excluded from the following analysis. This factor was included in the survey based on the previous findings that automobile commuters who ride their bikes on occasion are more likely to switch to electric bikes (de Kruijf et al., 2018).

3.1.4. Socio-demographics

Respondents were asked for general information that could be associated with their level of generating demand for travel such as age, number of children under 18, and household income. This study included questions on people's demographic traits that may affect a person's motivation to switch from driving to more environmentally friendly modes of transportation according to past literature. For example, younger individuals are more likely to utilize shared bikes (Y. Li et al., 2019). A study on California commuters found that having children and a higher level of education are associated with being a transit rider (Y. Lee et al., 2022).

3.1.5. Location

We asked the respondents about their home zip codes to filter out responses not from the application area.

While other factors, such as life stages or environmental awareness, can have an impact on people's mode choices, not all associated variables were included in the survey (Bai et al., 2020). Rather, in this study, we primarily focused on people's lifestyles, trip characteristics, and demographics in order to keep it from being overly long, to encourage people to fill out the whole survey and narrow the scope of this research.

3.2. Descriptive statistics of the sample

Table 1 presents the gender, age, and age distribution of the survey respondents to determine whether the sample is representative of Los Angeles residents. The gender and income distributions of the participant population are comparable to those of the 2020 Los Angeles County Census statistics, but there is a higher percentage of participants who are between the ages of 18 and 44 (U.S. Census Bureau, 2021).

4. Methodology

4.1. Non-negative matrix factorization to segment target audience

In this study, we employed non-negative matrix factorization (NMF) to segment the target audience and find their personas whose best option is SOV, but the second-best option is public transit, micromobility, carpool and none. A persona can refer to the public picture of one's personality, the social position that one adopts, or a fictional character, depending on the circumstances (McGinn & Kotamraju, 2008). In the case of this study, personas refer to different types of individuals that are more likely to perform a mode shift for a certain type of trip.

A Finnish group of researchers first proposed the NMF approach in the 1990s for dimensionality reduction (Paatero et al., 1991). In the transportation domain, many studies have utilized NMF to identify and interpret traffic patterns (Chen et al., 2018; X. Ma et al., 2020). In the marketing domain, NMF has also been utilized to understand individual behavior patterns. Clustering individuals based on their interests and other characteristics and building their personas for target audience marketing purposes is one of the NMF applications.

Each respondent was asked about their three most frequent trips, as well as trip characteristics and mobility choice preferences, in the survey (i.e. best option and second-best option). Based on the data, some people were discovered to be motivated for one type of trip but not for another. This showed that both individual and trip characteristics might influence people's likelihood of mode shift. As a result, decomposing the matrix containing both individual information and the trip characteristics for which they are likely or not likely to shift mode can reveal which types of people (e.g. demographics, lifestyles, bike ownership) are more likely to use a particular second-best mobility option for which trip (e.g. types of trips, trip duration). This study used the scikit-learn package to implement the method on the survey dataset in Python.

The following format is used to present the resulting personas:

- Individual-Trip type: Individual and Trip characteristics
- Second-best mobility option

First, a matrix, "X", was built, representing individual-trip types G_1, $G_2, \ldots$, G_g and the four second-best mobility options M_1, M_2, M_3, M_4, where M_1 = public transit, M_2 = micromobility, M_3 = carpool, M_4 = none (do not accept any other options than SOV). Individual-trip type is a unique combination of (G_1, $G_2, \ldots$, G_g) represented by g variable vectors. For example, an individual-trip type can be represented by three vectors ($g = 3$) which are age, gender and trip duration where age and gender are two variables that describe individual characteristics and trip duration is the one variable that describes trip characteristics, thus a unique individual-trip type could be (age = 18-34, gender = women, trip duration = less than 30 min) and a different type could be (age = 35-44, gender = men, trip duration = less than 30 min). The elements in the matrix X, represented by X_{ij}, are any values representing the number of individuals' trips that belong to the individual-trip type G_i $(i = 1, \ldots g)$ that chooses M_j $(j = 1, 2, 3, 4)$ as the second-best mobility option. For example, if 10 women belong to the age group 18-34 who are likely to shift to carpool for trips that are less than 30 min, then for individual-trip type (age = 18-34, gender = women, trip duration = less than 30 min), M_j =carpool and X_{ij}=10.

Once the matrix X is created, the number of distinct latent patterns can be discovered by decomposing it. The process of the non-negative matrix decomposition process is illustrated in Figure 1. X is the matrix of g individual-trip types (G_1, $G_2, \ldots$, G_g) and k second-best mobility options $(M_1, M_2, M_3, \ldots, M_k)$. Here $k = 4$ as this study is only interested in the four second-best mobility options which are public transit, micromobility, carpool, and none. NMF approximately factors matrix X into two matrices, U and V. When X is decomposed, U is a n^*p matrix and V is a p^*k matrix. In this case, n is the total number of unique individual-trip types; k is the four second-best mobility options; p is the number of latent patterns (personas of the target audience) which can control the number of the personas or

Table 1. Descriptive statistics of the survey respondents.

	Proportion
Gender	
Woman	45.1%
Man	48.5%
Non-binary	6.4%
Age	
18–34	46.3%
35–44	27.5%
45–54	17.3%
55–64	6.7%
65+	2.2%
Income	
$1 to $9,999	6.0%
$10,000 to $24,999	13.4%
$25,000 to 49,999	20.9%
$50,000 to 74,999	23.5%
$75,000 to 99,999	17.5%
$100,000 to 149,999	10.6%
$150,000 and greater	8.1%

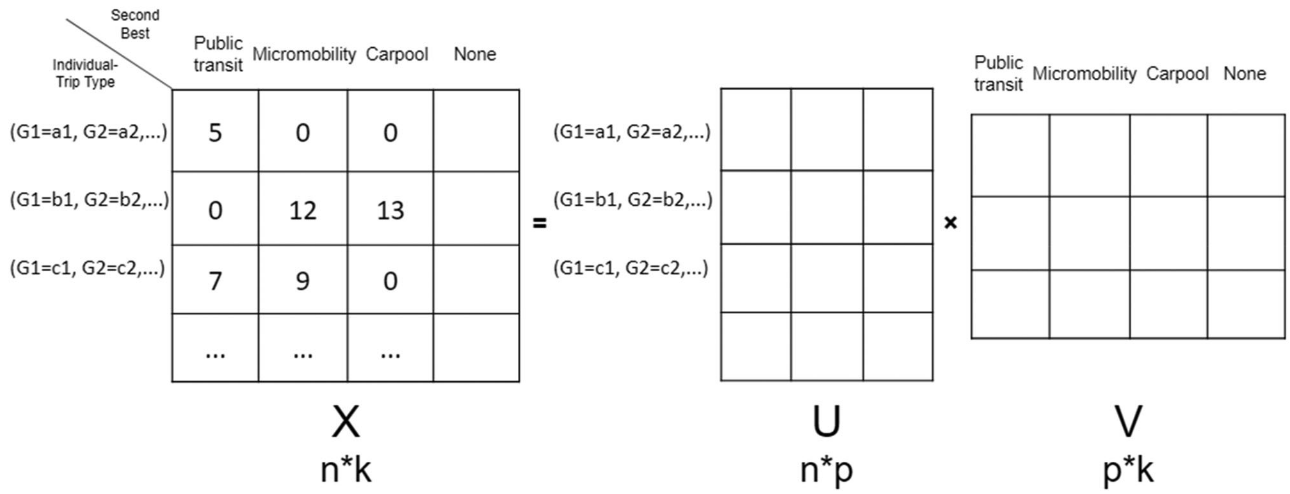


Figure 1. Illustration of NMF.

clusters output by the algorithm. The interpretation of U is that each column is a basis element, which is the fundamental building blocks from which we can reconstruct approximations to the original matrix X. The interpretation of V is that each column gives the coefficients of a row in the basis U. In other words, it tells how to reconstruct an approximation to the original matrix from a linear combination of the building blocks in U. In our case, the column in U represents the weight of each individual-trip type that contributes to the persona and the row in V represents how likely a persona, which is constructed by the columns with non-zero weights in matrix U, would choose a certain second-best mobility option.

4.2. Trip, demographic, and lifestyle clustering

Rather than directly using all the related variables collected from the survey variables for segmentation, the first step of this research was to identify distinct trip, demographic, and lifestyle patterns through clustering. This process helped find meaningful types of trips, demographics, and lifestyles that exist in the data. Furthermore, clustering beforehand may reduce the number of variables needed to describe the individual-trip type as some variables might be highly related (e.g. activity duration and departure time of day).

The K-Means clustering approach is well-known for its ability to divide large data sets into clusters. It has previously been used in the literature to cluster travel behavior using input variables such as land use features, trip reasons, and demographics, among other characteristics (Bao et al., 2017; Baumgarte et al., 2021; Jiang et al., 2012). Therefore, we utilized K-Means clustering in this study to identify the unique types of trips, demographics, and lifestyles in the data. The cluster ID from the clustering results (instead of the original trip-related variables) was used to describe the trip, demographic, and lifestyle characteristics. This helps reduce the influence of noise and outliers because each sample of the variables can be considered as being replaced by the cluster center.

We used six variables to understand individuals' trip characteristics including trip frequency, departure time flexibility, sociability, activity duration, departure time of day, and departure day of week. In order to fully understand the heterogeneity of the characteristics of the trips, trip purpose is not included in the clustering variables; for example, dining trips may be similar to social trips in terms of departure time flexibility and trip frequency, so the clustering algorithm considers them in the same cluster regardless of their different purposes.

Demographic variables describe individual characteristics. We used six variables to describe individuals' demographic characteristics. They were age, gender, household income, education level, number of dependents under 18, and job category.

Lifestyle variables also describe individual characteristics. Lifestyle characteristics represent the composition of different types of trips in one's day-to-day travels. For example, the lifestyle of those whose most frequent trip is commuting might be different than that of those whose most frequent trip is grocery shopping. We used three variables to describe individuals' lifestyle characteristics. They were the purpose of the most frequent trip, the second most frequent trip, and the third most frequent trip.

In addition to the trip, demographic, and lifestyle clusters, the input matrix of non-negative matrix factorization (NMF) includes bike ownership and trip duration respectively. We used bike ownership to examine if those who already own a bike are more likely to use micromobility or other shared means of transportation. Trip duration is distinct from other trip data because people who take the same type of trip (e.g. a low-frequency trip such as a leisure or social trip) may have various trip durations. Given the dataset, an individual-trip type is defined as a unique combination of factors, including their demographic cluster, lifestyle cluster, bike ownership, trip cluster, and trip duration.

4.3. Summary of target audience segmentation framework

The framework for identifying and segmenting the target audience for TDM programs is summarized in the following section. The potential target audience might not be just employees (e.g. volunteer groups, parents of schoolchildren, those who go to live sporting events). The procedures below are recommended to facilitate transportation planners, marketing specialists, and TDM programmers in reproducing this process for their state or region.

1. Data collection
   (1.1) Collect data about the independent variables such as demographics, lifestyles, and activity patterns. Other variables such as pro-environmental attitudes can also be collected.
   (1.2) Collect data about the identified dependent variable of interest, such as the likelihood for mode shift or actual mode switching behavior. Although the dependent variable in this study was mode shift motivation, which was obtained from a survey, this analysis framework can also be used to analyze data from mobile apps, where the dependent variable would be actual mode shift behavior.
2. Data processing
   (2.1) Apply the K-Means clustering method to the dataset to reduce the number of dimensions in the raw data. In this study, Python was used for processing the raw dataset, and the "sklearn.cluster.KMeans" package was used for implementing the K-Means clustering method.
3. Non-negative matrix factorization modeling
   (3.1) Apply the non-negative matrix factorization method to identify the distinct types of people from the processed data. In this study, the "sklearn.decomposition.NMF" package was used for implementing this method.
   (3.2) Summarize the characteristics of each type of the target audience.
4. Utilize the information of each type of target audience to develop personalized marketing campaigns and/or incentives that can then be applied on app-based or web-based platforms.

5. Results

5.1. Trip clusters

Three types of trips were identified by K-Means clustering algorithms: low frequency trip, high frequency trip and medium frequency trip. We discussed the characteristics of these three clusters in the following paragraphs. The characteristics (trip purpose duration, trip frequency, sociability, departure time flexibility and temporal patterns) of the three clusters are shown in Figure 2a–c.

Cluster 1: Low frequency trips (37.5% of the trips) - The duration of these trips is short, but the level of sociability is high. Weekend afternoons and evenings are the most popular times for these trips. It is hypothesized that these could be social or recreational outings with friends or family members.

Cluster 2: Medium frequency trips (29.7% of the trips) - The duration time, departure time flexibility, and sociability are all low for these trips. These trips, which are most likely grocery and other errand-type trips, take place primarily in the afternoons and evenings from Monday to Saturday

Cluster 3: High frequency trips (32.8% of the trips) - Among the three clusters, these trips had the longest duration and the lowest amount of sociability. The amount of flexibility in the trip departure time is limited. The majority of these trips take place on weekday mornings and afternoons. It is hypothesized that these could be commuting (mainly on weekday mornings) or pick-up or drop-off trips (both weekday mornings and afternoons).

The distribution of trip purposes among these three trip clusters was examined and shown in Figure 3. The great majority of trips in the medium frequency cluster are grocery and errand trips (68%). The low frequency cluster is dominated by social and leisure visits (54%), but shopping and errand trips still make up a major fraction of the total (48%). The most common trips in the high frequency cluster are commuting and pick-up or drop-off trips (72%).

5.2. Demographic clusters

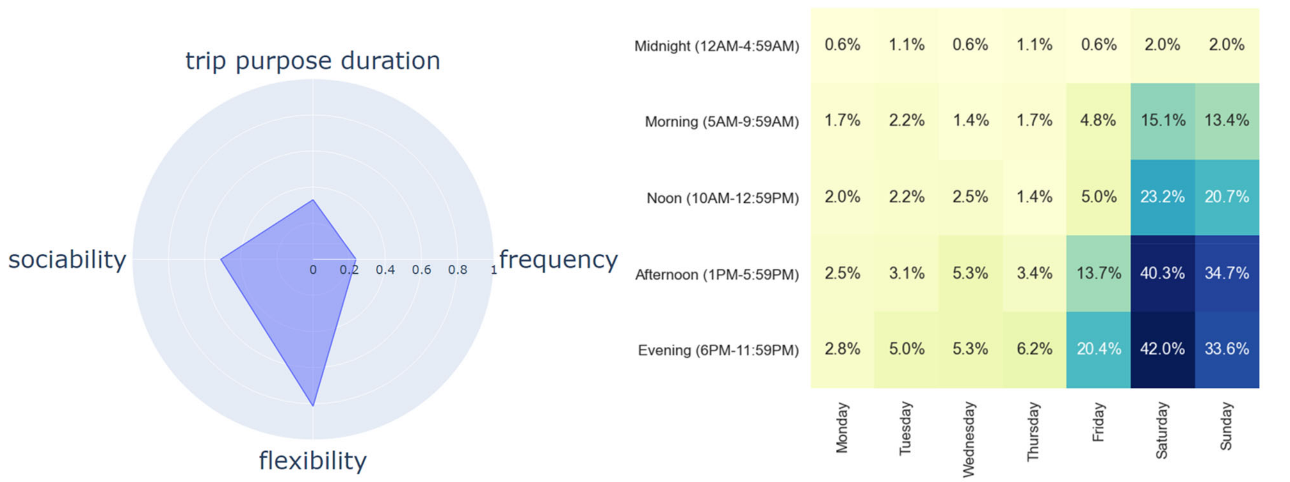
Four demographic clusters were identified from the K-Means clustering analysis. These include young low-income, young-middle-age middle-high-income, middle-age low-income women, and elder middle-income. Table 2 shows the age, gender, annual household income, education level, and job distribution of the four clusters. Only dominant categories are listed.

Cluster 1: Middle-age-elder middle-income (14.5% of the respondents) - Over 80% of the respondents in this cluster are over the age of 45 and 84.3% of the respondents do not have dependents in the household. Their income is about the median income in the Greater Los Angeles Area which is $68,044 (U.S. Department of Commerce, 2019), with 60.7% of the respondents' annual household income between $50K and 100K. About half of the respondents have college degree and 29.4% of them have high school degree.

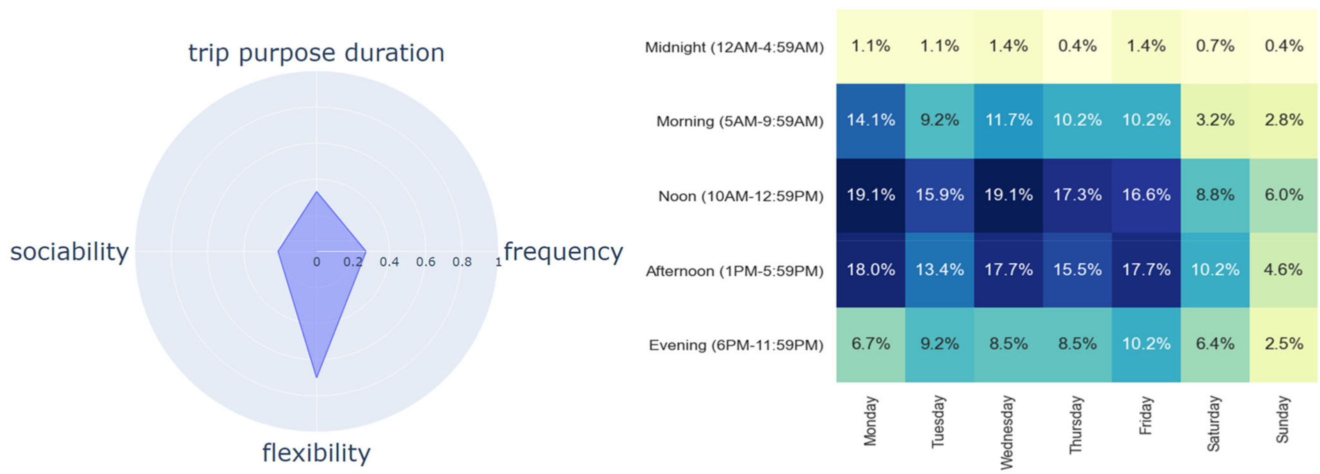
Cluster 2: Young-middle-age high income (34.8% of the respondents) – 89.2% of the respondents in this cluster are younger than 44. There are slightly more men than women in this cluster. 71.2% of the respondents have an annual household income higher than $100K. Their education level is relatively high, with 29.5% of the respondents having post-graduate degree.

Cluster 3: Young-middle-age middle income (17.9% of the respondents) - The age distribution is similar to that of cluster 2, with 85.7% of the respondents in this cluster younger than 44. The annual household income of this cluster is similar to that of cluster 1, with 76.1% of the respondents have annual household income between $50K and 100K. Their education level is also similar to that of cluster 1.

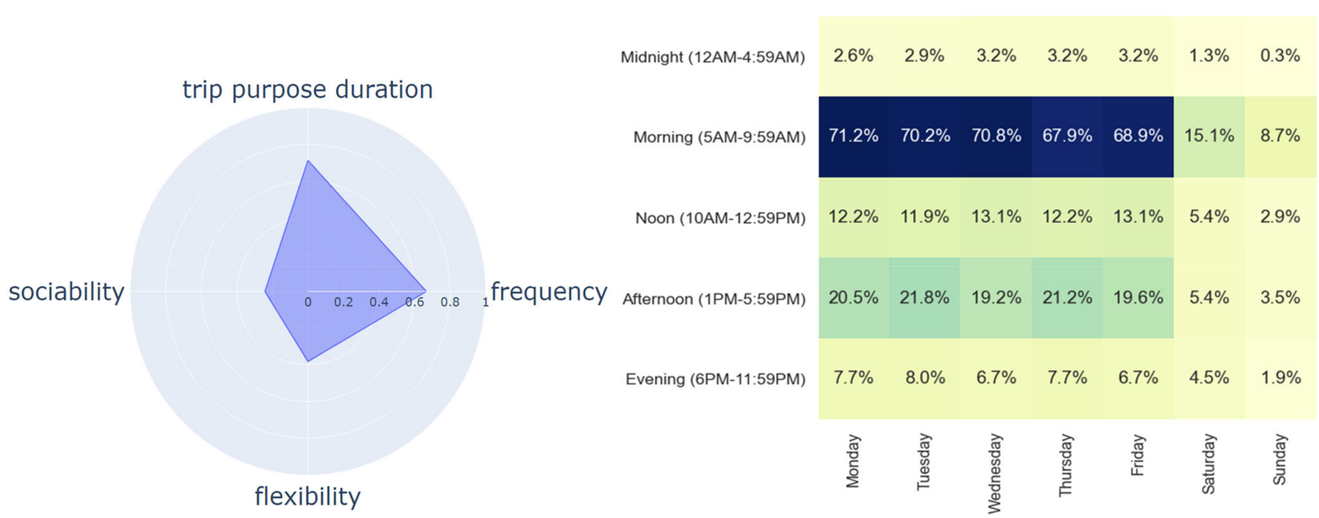
Cluster 4: Young low-income (32.8% of the respondents) – 75.6% of the respondents in this cluster are between the ages of 18 and 34. 73.9% of the respondents do not have dependents under the age of 18. 44.3% of the respondents have annual household income between $25K and $50K, below the median annual household income in the Greater Los Angeles Area.



(a) Low frequency trip



(b) Medium frequency trip



(c) High frequency trip

Figure 2. Trip clusters attributes (a) Low frequency trip (b) Medium frequency trip (c) High frequency trip.

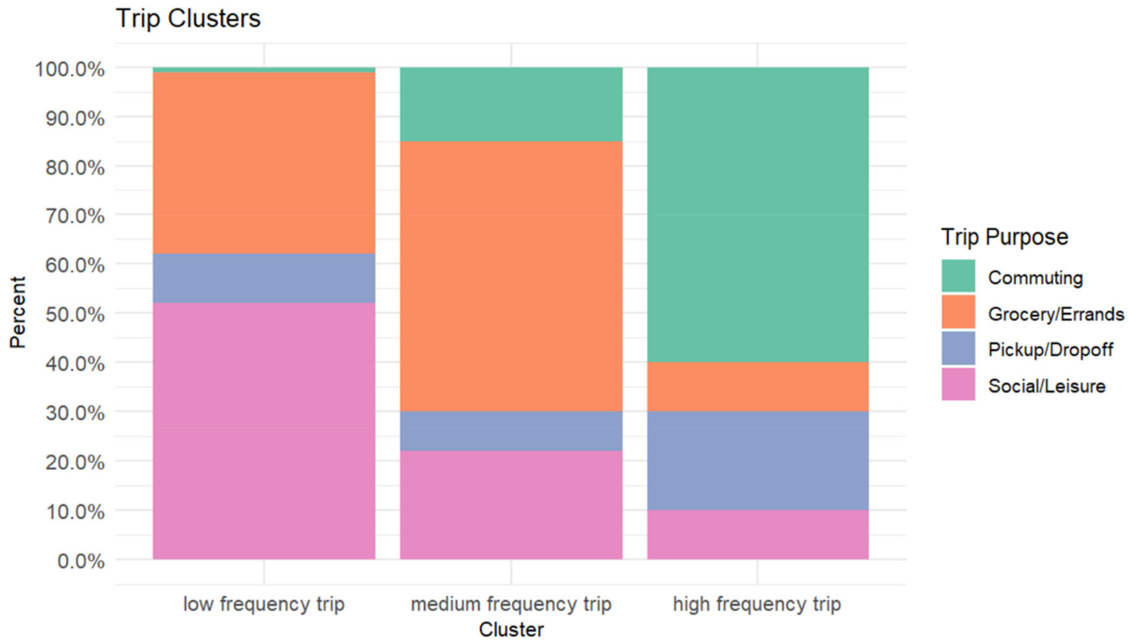


Figure 3. Trip purpose distribution of the three trip clusters.

Table 2. Demographics clusters.

Category\Clusters	Middle-age-elder middle income (14.5%)	Young-middle-age high-income (34.8%)	Young-middle-age middle-income (17.9%)	Young low-income (32.8%)
Age	45-54 45.0% 55-64 35.2% 65 + 19.6%	18-34 56.5% 35-44 32.7%	18-34 49.2% 35-44 36.5%	18-34 75.6% 35-44 23.4%
Gender	Woman 56.8%	Man 58.1%	Woman 50.7%	Woman 50.4%
Household income	\$50,000 to 74,999 33.3% \$75,000 to 99,999 27.4%	\$100,000 to 149,999 36.0% \$75,000 to 99,999 35.2%	\$50,000 to 74,999 55.5% \$75,000 to 99,999 20.6%	\$25,000 to 49,999 44.3% \$50,000 to 74,999 30.4%
# dependents under 18	0 84.3%	0 59.0% 1 22.9%	2 55.5% 1 23.8%	0 85.2%
Education level	College 50.9% High School 29.4%	College 53.2% Post College 29.5%	College 58.7% High School 23.8%	College 62.6% High School 21.7%
Job category	Education 17.6% Other 13.7% Business 11.7%	Business 15.5% Tech 15.5% Health 13.9%	Business 25.3% Education 14.2% Other 9.5%	Tech 15.6% Business 13.9% Other 11.3%

5.3. Lifestyle clusters

Three lifestyle clusters were identified by the K-Means algorithm and visualized in Figure 4: Commuters (most frequent trip is commuting), and people with flexible lifestyle (most frequent trip is grocery or errands). Figure 5a and b represents the distribution of the purpose of the most, second most, and third most frequent trip of each cluster.

Cluster 1: Commuters (most frequent trip is commuting) - This group accounted for 76.4% of all respondents. The majority of their most frequent trips are commuting trips. Their second and third most frequent trips are social or leisure trips, as well as grocery or errand trips.

Cluster 2: People with flexible lifestyle (most frequent trip is grocery or errands) - This category accounts for 23.6% of all responders. Most of their frequent trips are for groceries or errands, as well as pick-up or drop-off.

5.4. Matrix decomposition results

After the input matrix was decomposed, matrix *U* and V were visualized in Figure 6a and b. The numbered personas in Figure 6a and b were the clusters identified by the algorithm. One of the algorithm's input variables is the number of clusters. The number of clusters is set to four in our case since the algorithm is intended to discover the personas for which public transportation, micromobility, carpool, and none are the most likely second-best options. The column in U represents the weight of the coefficient of each individual-trip type that contributes to the persona and the row in V represents how likely a persona, which is constructed by the columns with non-zero coefficients in matrix U, would choose a certain second-best mobility option. The diagonal values in each row in the V matrix are higher than the other values in that row, indicating that the algorithm has well segmented the audience to each second-best option.

5.5. Personas summary

Each persona (i.e. those who are not likely to shift modes, likely to shift to public transit, micromobility, carpool) can have multiple individual-trip types. Thus, for each persona, the individual-trip type with the largest weights can be interpreted as the most impactful group for that corresponding pattern. The four

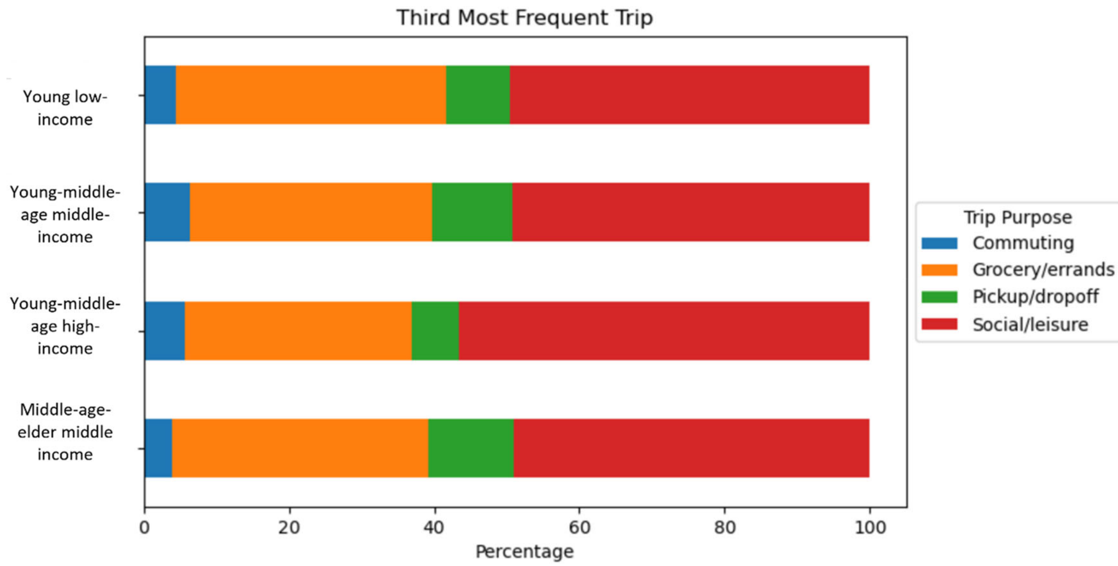


Figure 4. Third most frequent trip distributions of the four demographic clusters.

individual-trip types that have the highest coefficient in each persona were shown in the tables below. We summarized the key findings from these personas in the following sections.

5.5.1. Personas representing those who are not likely to shift modes

65.5% of the individual-trip types whose best option was SOV were segmented into this cluster representing those who are not likely to shift mode. Table 3 summarizes their personas. The weight in the second column represents the contribution of this individual-trip type to this persona. The individual-trip types with the largest weights are listed in Table 3 and can be interpreted as the most impactful groups for this persona. For young-middle-aged high-income individuals, long-distance (30–45-minute driving) social or leisure trips are less likely to shift mode.

5.5.2. Public transit personas

5.7% of the respondents whose best option was SOV were segmented into the public transit cluster. Table 4 summarizes the public transit personas. The weight in the second column represents the contribution of this individual-trip type to these personas. The individual-trip type with the largest weights is listed in Table 4. These can be interpreted as the most impactful groups for the public transit persona. Young-middle-age with middle or high income is more likely to use public transit for social and leisure purposes. Young people with low income are more inclined to use public transportation. Additionally, people are more likely to shift to public transit on short trips (0–30 min). Additionally, the findings reveal that people who own a bicycle are more likely to use public transit for commuting.

One of the public transit persona's shortcomings is that the public transit accessibility element, such as the distance from respondents' homes, is not taken into account while creating their personas. Because of privacy considerations, this type of precise location information is not appropriate to ask in the survey used in this research. The survey did, however, ask about home zip codes, which are used in the next section to analyze the qualitative association between the location distribution of the second-best options and transportation facilities.

5.5.3. Micromobility personas

4.6% of the individual-trip types whose best option is SOV have been segmented into the micromobility cluster. Table 5 summarizes the micromobility personas. The weight in the second column represents the contribution of this individual-trip type to these personas. The individual-trip type with the largest weights is listed in Table 5. These can be interpreted as the most impactful groups for the micromobility persona. Young-middle-age middle-income is more likely to be motivated and capable to use micromobility for short commuting trips. Middle-aged-elder middle-income and young-middle-age high-income are also more likely to use micromobility on social or leisure trips. This could be done for exercise or in the company of friends or family. Additionally, these individuals are also likely to own at least one bicycle. This is most likely due to the fact that they are more accustomed to bicycling and have easier access to it.

5.5.4. Carpool personas

24.2% of the respondents whose best option is SOV have been segmented into the public transit cluster. Table 6 summarizes the carpool personas. The weight in the second column represents the contribution of this individual-trip type to these personas. The individual-trip type with the largest weights is listed in Table 6. These can be interpreted as the most impactful groups for the carpool persona. Young-middle-age high-income people are more likely to carpool for short social or leisure trips (0-30 min), where they may feel more at ease than on a long-distance trip. They are mostly regular commuters, with commuting being their most common travel.

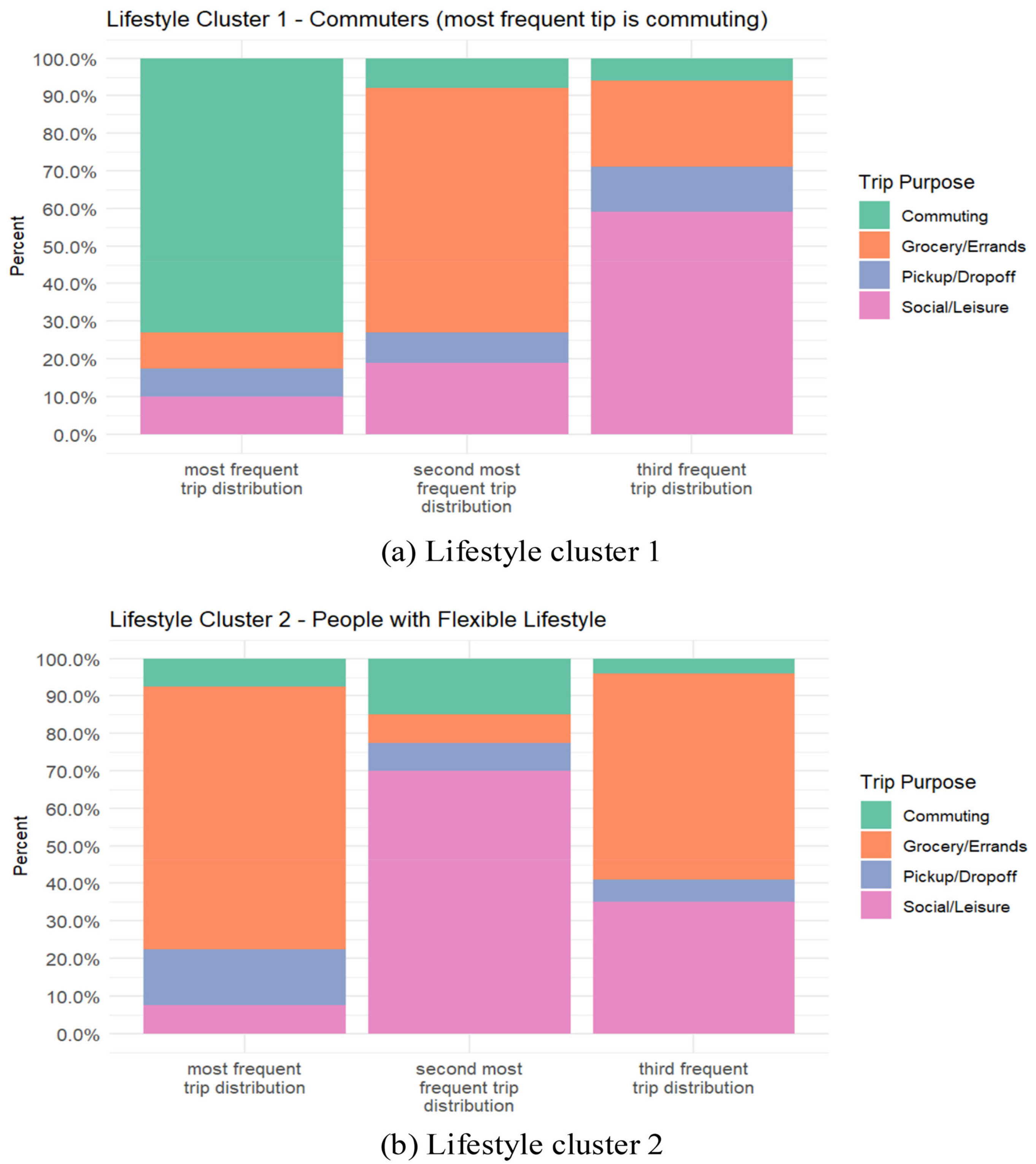


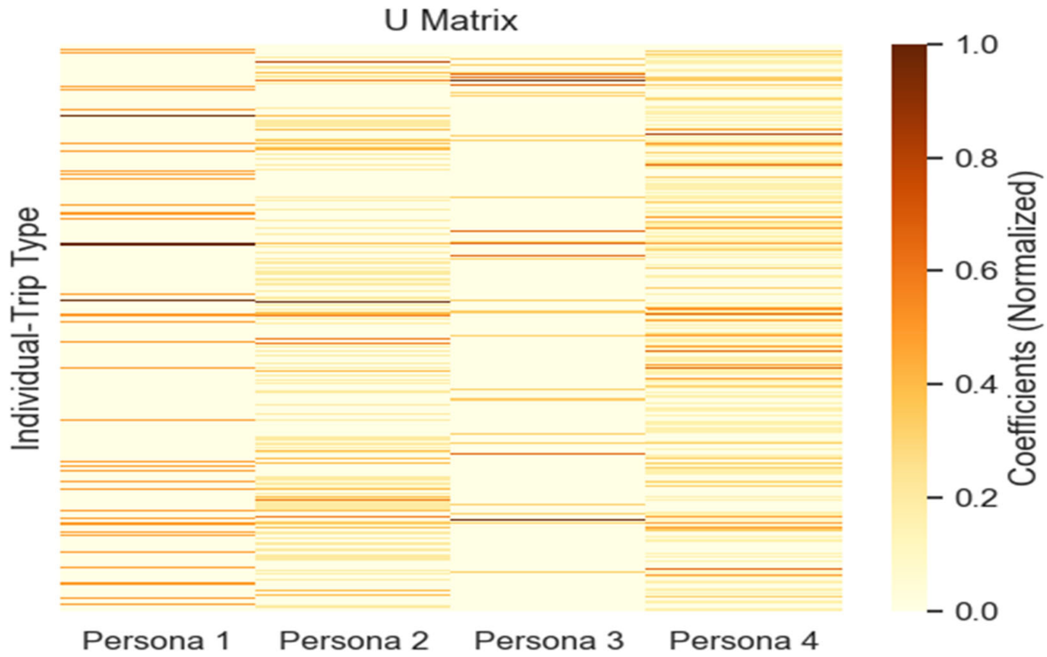
Figure 5. Trip distribution by lifestyle clusters (a) Lifestyle cluster 1 (b) Lifestyle cluster 2.

6. Discussion

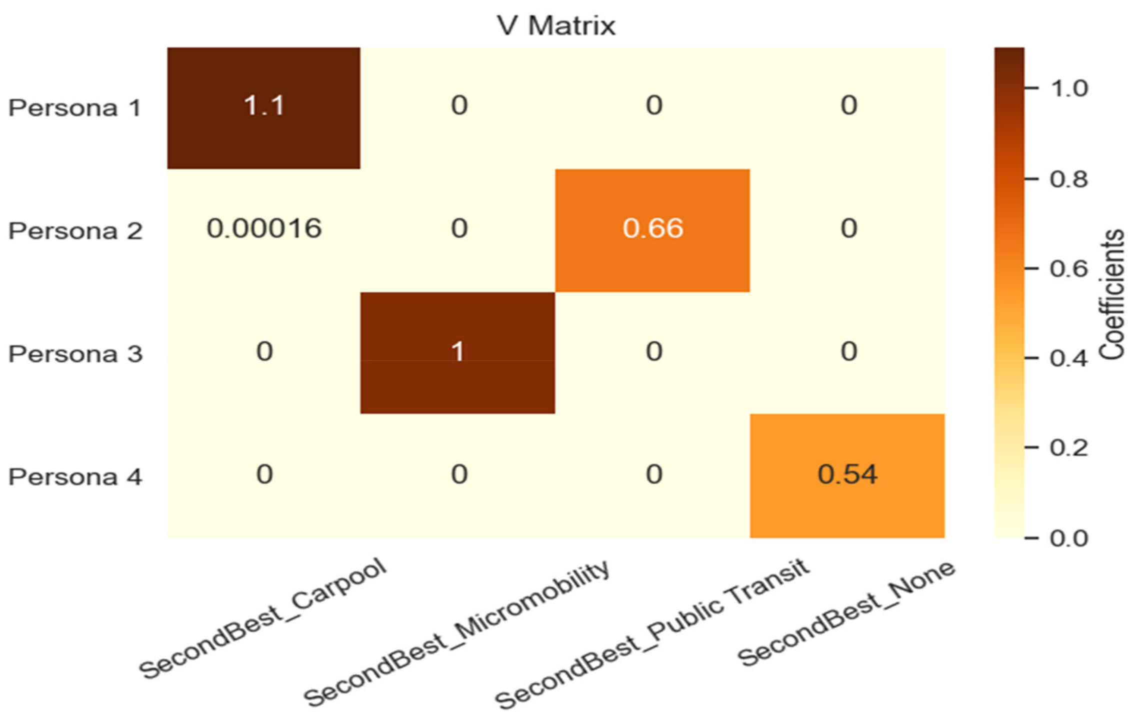
6.1. Relationship between demographic and lifestyles of the target audience

To find out why the four demographic groupings have different second-best mobility options preferences, a more in-depth investigation of the demographic characteristics and lifestyles of the target population was undertaken (i.e. young low-income, young-middle-age middle-income, young-middle-age high-income, and middle-age-elder middle-income).

The lifestyles (the composition of different types of trips in one's day-to-day travels) of the four demographic clusters were first examined. Figure 7 shows the most frequent trip distribution of the four demographic clusters. Individuals in the young-middle-age middle-income and middle-age-elder middle-income groups were found to take fewer commuting trips than those in the other two groups, with the former



(a) U Matrix



(b) V Matrix

Figure 6. Matrix Factorization Results (a) U Matrix (b) V Matrix.

having the highest proportion of pick-up or drop-off trips and the latter having the highest proportion of grocery or errand trips among the four clusters. Apart from commuting, young-middle-age middle-income people are more likely to take on more family responsibilities than people in other clusters of picking up or dropping off their family members.

Figure 8 shows the second most frequent trip distribution of the four demographic clusters. Grocery or errand trips are the most common second most frequent trip in all

Table 3. Personas representing those who are not likely to shift.

Persona ID	Weight	Trip Cluster	Travel time	Lifestyle	Demographic	Bike Ownership
None_1	11.093	Medium frequency trip (grocery)	45-60 min	Commuter	Young-middle-age high-income	0
None_2	9.266	Medium frequency trip (grocery)	15-30 min	Flexible lifestyle	Middle-age-elder middle income	1
None_3	7.395	Low frequency trip (social/leisure)	45-60 min	Commuter	Young-middle-age high-income	0
None_4	7.395	Low frequency trip (social/leisure)	45-60min	Commuter	Young-middle-age high-income	1

Table 4. Public transit personas.

Persona ID	Weight	Trip Cluster	Travel time	Lifestyle	Demographic	Bike Ownership
Public transit_1	3.057	Low frequency trip (social/leisure)	0-15 min	Commuter	Young-middle-age middle-income	1
Public transit_2	3.048	Low frequency trip (social/leisure)	15-30 min	Flexible lifestyle	Young-middle-age high-income	2
Public transit_3	3.048	High frequency trips (commuting)	0-15 min	Commuter	Young low-income	2
Public transit_4	3.048	High frequency trips (commuting)	0-15 min	Commuter	Young low-income	1

Table 5. Micromobility personas.

Persona ID	Weight	Trip Cluster	Travel time	Lifestyle	Demographic	Bike Ownership
Micromobility_1	2.973	High frequency trips (commuting)	0-15 min	Commuter	Young-middle-age middle-income	3
Micromobility_2	1.982	High frequency trips (commuting)	0-15 min	Commuter	Young-middle-age middle-income	2
Micromobility_3	1.982	Low frequency trip (social/leisure)	0-15 min	Flexible lifestyle	Middle-age-elder middle income	2
Micromobility_4	1.982	Low frequency trip (social/leisure)	0-15 min	Commuter	Young-middle-age high-income	2

Table 6. Carpool personas.

Persona ID	Weight	Trip Cluster	Travel time	Lifestyle	Demographic	Bike Ownership
Carpool_1	4.575	Low frequency trip (social/leisure)	15-30 min	Commuter	Young-middle-age high-income	0
Carpool_2	3.661	Low frequency trip (social/leisure)	0-15 min	Commuter	Young-middle-age high-income	0
Carpool_3	2.749	High frequency trips (commuting)	15-30 min	Commuter	Young-middle-age high-income	1
Carpool_4	2.749	Low frequency trip (social/leisure)	15-30 min	Flexible lifestyle	Young-middle-age middle-income	0

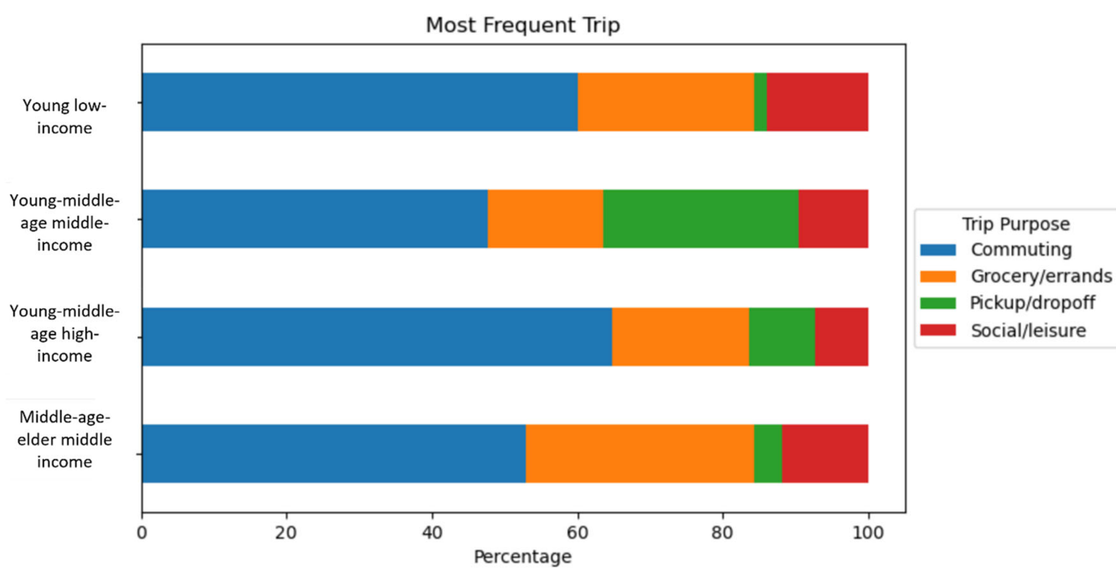


Figure 7. Most frequent trip distributions of the four demographic clusters.

clusters. Young-middle-age high-income have more social or leisure trips and slightly more commuting trips as their second most frequent trips than other clusters of people. These people are more likely to participate in social activities related to their work or spend their free time with friends or family.

Figure 4 shows the third most frequent trip distribution of the four demographic clusters. Social or leisure trips are popular as people's third most frequent trips among all demographic clusters.

Second, the demographic distribution of the two lifestyle clusters was also investigated. Two clusters of lifestyles were identified from the analysis. One is the lifestyle of a commuter whose most frequent trip is commuting, and the other lifestyle is more flexible, which could be because they have more flexible commute patterns or because they have more family commitments.

The age (Figure 9), gender (Figure 10), number of dependents under 18 (Figure 11), and job distribution (Figure 12) among the two clusters were further examined.

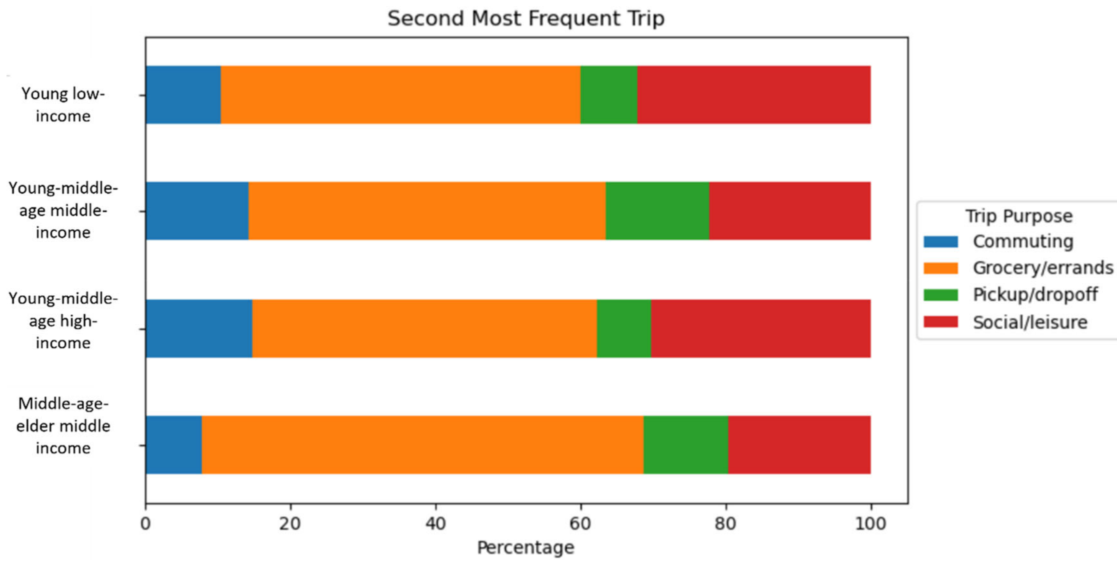


Figure 8. Second most frequent trip distributions of the four demographic clusters.

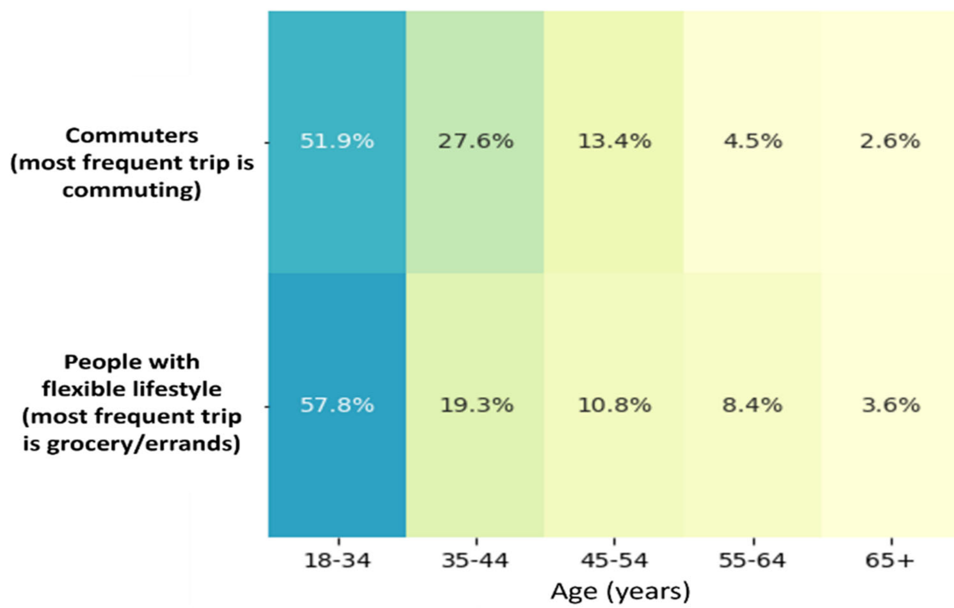


Figure 9. Age distribution of two lifestyle clusters.

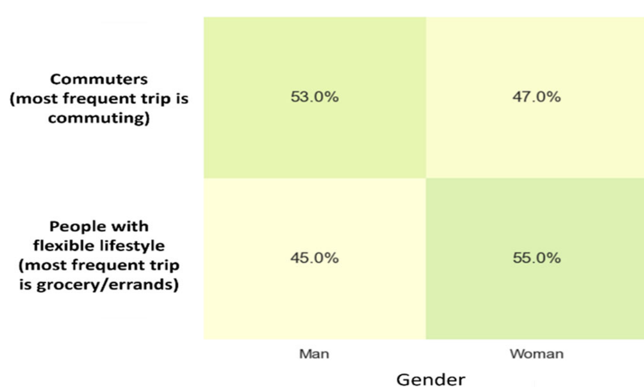


Figure 10. Gender distribution of two lifestyle clusters.

The findings reveal that commuters whose most frequent trip is commuting (Lifestyle Cluster 1) are slightly more likely to be men (53%) and to be between the ages of 18 and 34 (51.9%) and 35 and 44 (27.6%). Nearly 60% of commuters have no dependents under the age of 18, while 40% have one or more dependents under the age of 18. Business is the most common job category in this group. People in this cluster are more likely to have a relatively fixed working time and location, thus their time flexibility is relatively low for most of their trips.

Those who have a relatively flexible lifestyle (Lifestyle Cluster 2) are more likely to be women (55%). They have a higher percentage of respondents between the ages of 18 and 34 (57.8%) and between the ages of 55-64 (8.4%) than commuters whose most frequent trip is commuting. Those who have a relatively flexible lifestyle are also more likely to have no dependents under the age of 18 than commuters whose most frequent trip is commuting. Unlike commuters whose most frequent trip is commuting, those who have a relatively flexible lifestyle include a significant ratio of people

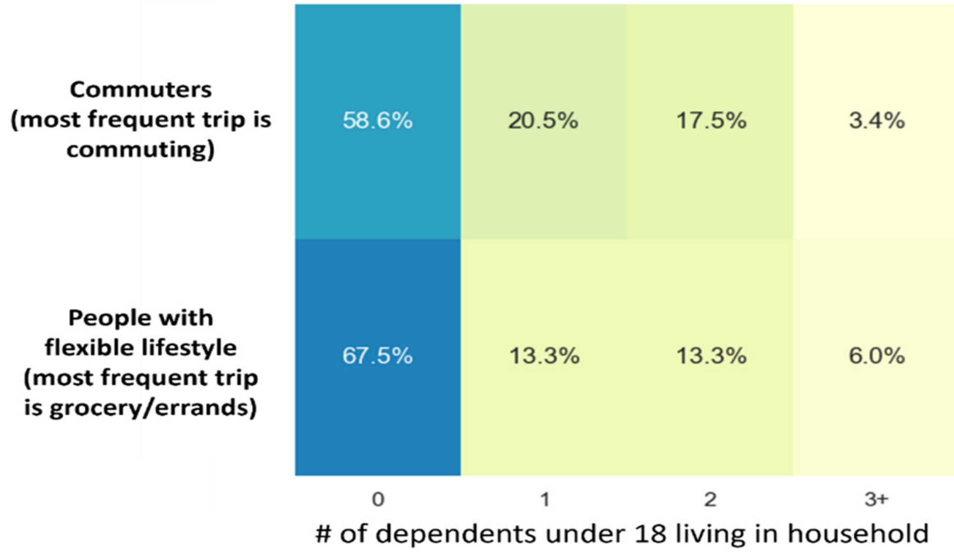


Figure 11. Number of dependents distribution of two lifestyle clusters.

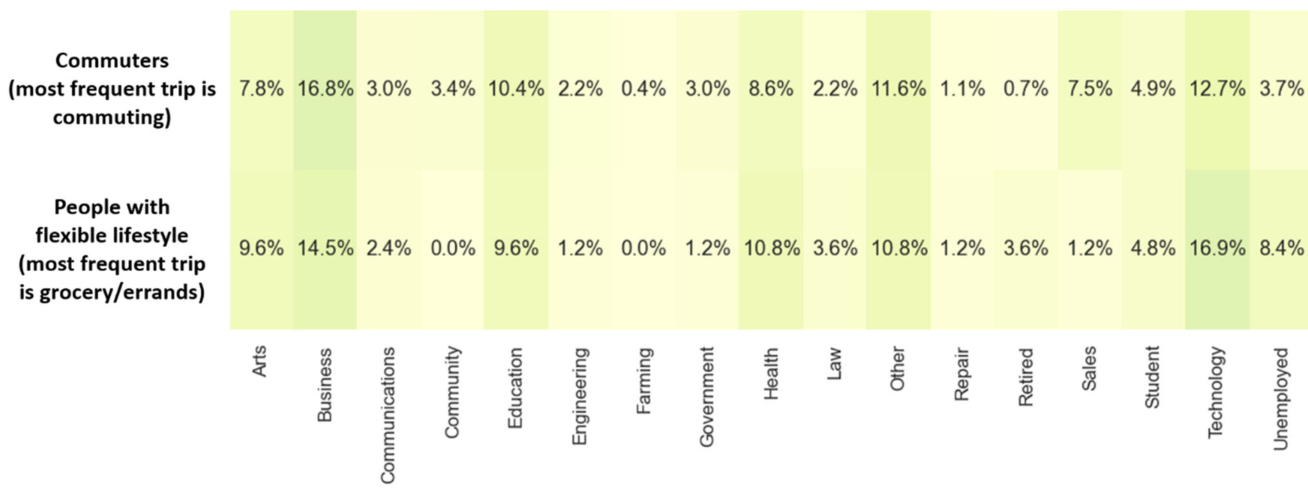


Figure 12. Job distribution of two lifestyle clusters.

working in the technology industry, retired or unemployed, which could indicate that they have more time flexibility.

In summary, the findings reveal that young-middle-age middle-income people are more likely to take on more family responsibilities, as they have the highest percentage of individuals whose most frequent trip is pick-up or drop-off. This could explain why young-middle-age middle-income people who live a commuter lifestyle are less likely to use public transit or carpool for their most frequent trips which might be commuting, pick-up or drop-off trips, or both on the same trip, making the mode shift inconvenient as these trips have relatively lower time flexibility.

6.2. Key findings and interpretation

In this study, we identified the primary characteristics of the target audience through the audience segmentation process using the non-negative matrix factorization. The findings are summarized and discussed in the following list. This study advances the understanding of why some population segments are more (or less) inclined to change their mode choices than others by examining the connections between people's lifestyle patterns, demographics, and trip characteristics. Transportation agencies can use the proposed clustering and audience segmentation approach to create personas of those who are and are not likely to switch modes to construct more successful and targeted TDM programs.

Many of the findings in this paper appear intuitive; however, it is critical to demonstrate that research results exist to support these hypotheses as well as discuss the policy implications beyond these findings. This research finding has not been demonstrated in recent literature, to the authors' knowledge. Without this information, transportation professionals are unable to make informed policy and program decisions.

- Young-middle-aged people with a high income or those who are elderly are more resistant to TDM programs that encourage the use of the alternative mobility options, particularly when it comes to long-distance travel, which is likely due to a high perceived value of time and avoidance of situations that are inconvenient or uncomfortable, such as carrying multiple grocery bags.

- Young people with low income are more likely to use public transit for commuting trips, which is likely due to the availability of a transit pass or the desire to save money on gas and parking. Additionally, those who have a relatively higher income (young-middle-age middle-income and young-middle-age high-income) who have a flexible lifestyle are more likely to use public transit for social or leisure trips than commute trips. Based on the job distribution, those who have a relatively flexible lifestyle include a substantial ratio of those working in the technology industry, retired or unemployed, which could indicate that they have more time freedom than commuters. As a result, they are likely to have greater time flexibility which gives them some time to try modes they are not that familiar with than commuters. Furthermore, people are more likely to shift to public transit on short trips (0-30 minutes). This is probably because public transit can provide a relatively smoother ride than driving without adding much time to the trip. The results regarding job characteristics support researchers' and practitioners' understanding of how TDM programs can be created to suit traffic demands given the flexible work schedule policies, which are more common following the pandemic, where people do not have to commute to the office every workday.
- Those who have a relatively higher income (young-middle-age middle-income and young-middle-age high-income) are more likely to shift to micromobility and carpool for a short commuting trip. This is likely due to the fact that micromobility and carpooling provide them with more predictable travel times than public transit. Because the trip is relatively short, it may also provide a smoother experience than driving through congested areas without adding too much time to the total travel time. Furthermore, people were found to be more likely to carpool for social and leisure trips. This is also consistent with the past literature that carpooling was more frequently used for leisure and shopping trips than other types of trips (Gheorghiu & Delhomme, 2018).
- Those whose household owns at least one bike are more likely to select micromobility or public transit as their second-best mobility option. This result suggests that having a bike may have a significant impact on one's willingness to participate in TDM programs, as these people are at least familiar with one sort of non-personal mode of transportation and may be open to other shared modes of transportation. This is also consistent with past literature indicating that those who own a bike are more likely to switch from a car to their own bike or other shared modes (de Kruijf et al., 2018; Song et al., 2019).
- Commuting and social or leisure trips are preferable trips to promote mode shift than grocery trips. This could be because using carpool, public transit, or micromobility, which sometimes includes some walking and waiting, is inconvenient for people who have a lot of bags and/or who have dependents (Rokseth et al., 2021). Past TDM programs mainly focused on encouraging mode shift in commuting trips (UCR School of Business Center for Economic Forecasting & Development, 2020). As more trips, such as social or errand trips, increase following the pandemic, the results regarding trip purposes can help researchers understand how they can develop TDM programs to encourage mode shift for different trip purposes. Given the various trip characteristics associated with various trip purposes, such as departure time flexibility, TDM programs could be developed to be more targeted, making them more acceptable to and used by more people.

6.3. Policy implications

Based on the findings, this research offers several implications for transportation agencies in their development and marketing of mode shifts or other transportation behavior change programs. Given the different individual and trip groups discovered for various modes of transportation, it appears that only some people are driven to switch from driving to other modes of transportation. Given this, when creating a behavior change program, this research suggests that the first step should be to identify the target audience. This study presents an effective audience segmentation and persona construction method to help transportation agencies better understand different types of people that are more likely to participate in these programs, which, in turn, will inform their policies and practices.

Following the identification of the target audience, the research findings suggest that the second-best step is to identify what targeted information and incentives should be delivered to the target audience to influence them toward choices that improve their personal mobility experiences and the congestion, air quality, and climate goals. To accomplish this, surveys and focus groups could be developed to understand the barriers and pain points of these personas, such as why or under what conditions they intend to convert from driving to other modes of transportation, and what are the factors that hinder them from doing so. For example, the Los Angeles Metro system offers a variety of carpool and vanpool services, such as RideMatch, which links commuters who have the same travel time and route and are looking to save money. The characteristics of the carpool personas identified in this study could be used to identify the program's target audience and offer targeted campaigns or follow-up surveys to learn more about their needs and barriers to using the service, potentially resulting in a higher rate of engagement and participation.

As a result, not only will agencies be able to improve their communication strategies and better reach out to their target audiences, but they will also be able to strengthen their physical infrastructure in order to deliver better services to them. Personalized behavioral interventions may be devised to target different personas, allowing for more behavior change to occur and to endure longer, which is potentially more cost-efficient than traditional one-size-fits-all strategies (Yuan et al., 2021).

6.4. Limitations

There are some limitations to this research. First, due to data collecting limitations, the analysis was limited to young and middle-aged people. According to research that looked at the demographics of Amazon Mechanical Turk respondents, over 60% of them are under the age of 40, making them much younger than the general population of the United States. As a result, other channels for distributing surveys could be employed to provide a more thorough analysis of the older generations. It is also worth noting that we may have biased the survey respondents by providing them with a set of mobility options, as the respondents might want to try other options such as ferry or on-demand microtransit as well. Second, a future research step that includes a validation process would be beneficial to better understand the real impact of this methodology. For example, a real marketing campaign could be run to assess whether the study's target demographic is more likely to engage with the advertising and participate in the mode shift activities. Third, because this survey was completed during COVID-19, there is a potential that those who did not consider a second-best option would be more open to more sustainable alternatives such as public transportation in the near future. Additional research could be conducted before commencing a transportation demand management program to have a better picture of the target audience for a specific region. The transferability of the results may be constrained in terms of directly assisting other regions in identifying the target audience for their TDM programs as this study focused on a particular region with unique traffic characteristics. However, the general framework, made up of trip and individual clustering and a matrix factorization approach, was proven to be an effective method to segment target audiences and can be applied in various regions, such as cities in Europe or Asia. Given each individual region's unique mobility patterns, transportation agencies may want to include additional questions in their survey. For example, when using this framework in places where electric bikes and scooters are reasonably common, more questions may need to be asked about these mobility options. Furthermore, future research may examine the variables such as age and gender that might affect people's long-term mobility patterns under the influence of different TDM strategies. Last but not least, we primarily focused on identifying the target audience (i.e. both motivated and capable of using the alternatives) for TDM programs; thus, future work should further investigate those who are not likely to shift modes and why (e.g. unwilling to try the alternatives, unaware of the alternatives, motivated to use but not able to access the alternatives). Understanding these groups of people is critical in assisting transportation agencies in better serving their various requirements or developing more targeted strategies to change mode use behavior in their communities.

7. Conclusions

In this study, we proposed the approach of finding the target audience for transportation mode shift programs in order to improve the program's effectiveness and assist the development of more personalized solutions. We used a matrix factorization algorithm to identify and segment target audiences for transportation mode shift programs and identify meaningful audience segments for potential public transit, micromobility, and carpool 'switcher'.

The proposed approach can assist in performing user analysis when TDM programmers, marketing specialists, and transportation planners are planning and preparing for the TDM programs. The findings can assist in the creation of more tailored TDM strategies, which can lower overall program development costs and enhance behavioral impact. The following bullet points highlight the significant findings pertaining to the proposed methodology:

- This method can facilitate the design of a survey to understand the relationship between mode shift motivation and individual characteristics of a certain state or region.
- This modeling framework provides a method to identify target audience characteristics from high-dimensional behavior data.
- The results from the audience segmentation process can facilitate the design of individualized marketing campaigns, not only for employers, but can also be for state and regional agencies, primary schools, universities, communities, and special events.

There were several takeaways from the research results that transportation agencies can use when developing target audiences for incentives or other programs to encourage mode shift to more sustainable mobility options. The following bullet points highlight the significant findings:

- For short-distance leisure or social trips, young-middle-aged middle-high-income people are more likely to shift to public transit, carpool, or micromobility mobility options from SOV.
- Young-low-income people are more likely to shift to public transit on short-distanced commuting trips.
- People who live a flexible lifestyle, work in the technology business, are retired, or are unemployed are more likely to switch to public transit.
- Young-middle-aged middle-income people are more likely to have pick-up and drop-off responsibilities, making them less likely to switch modes.
- In general, commuting and social or leisure trips are preferable trips to promote mode shift than grocery trips.
- Those whose household owns at least one bike are more likely to switch to a micromobility or public transit mode than those who do not.
- Young-middle-aged people with a high income and older adults are more resistant to TDM programs, particularly when it comes to long-distance travel.

These research findings and development methodology provide critical insight into what type of people are more likely to shift from SOV to more sustainable mobility options. The developed methodology was successful at determining those who are more or less likely to shift mode and, thus, can be directly employed by transportation agencies for specific regions, allowing them to investigate their own target audiences. This can aid in the design of future transportation marketing and behavior change programs and policies for more sustainable, climate-conscious transportation systems, which are imperative to reducing global emissions from the transportation sector.

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